

The gate runs, and nine of its ten checks provably enforce. First round with no critical finding.

Sixth independent pass. R5-1 is fixed, so I could finally exercise the smoke gate's accept path — then mutate it one field at a time to prove each check individually. One check is inert.

FLOW_CLEAR_WITH_WARNINGS

No critical findings and no bypass landed. Three items remain: one newly-found inert check, the preflight runtime regression from round 5, and the governance question open since round 2.

CRITICALS	GATE CHECKS PROVEN	NEW WARNINGS	CARRIED FORWARD	TESTS
0	9 / 10	1	3	131

EXECUTIVE VERDICT

The first clean round, and why the verdict changed

R5-1 is fixed. `repo_root` and `hashlib` are now module-level in `collect.py`, and for the first time in this series the smoke gate's accept path executes: a genuine acceptance passes all ten checks and reaches engine construction. That unblocked the test I have been unable to run — mutating the genuine acceptance one field at a time to prove each newly-reachable check actually rejects.

Nine of ten do. Validator SHA, violation count, coverage percentage, rows-examined consistency, and run-directory existence each blocked with a distinct, correct error. The tenth — `exact_timestamp_equality_verified` — is inert, for a reason worth reading carefully (R6-1). It is mitigated: the substantive claim it duplicates is enforced unconditionally two lines above it.

I am moving the verdict to **FLOW_CLEAR_WITH_WARNINGS**. Across six rounds the framework now mechanically prevents the failure classes the brief asked about: look-

ahead is measured rather than asserted and independently confirmed at 2,002/2,002 exact; audited code cannot change without the seal going stale across a 53-file AST closure; DEV/OOS cannot be reached, including through warmup; a full run cannot proceed without a validated, seal-bound smoke; and exact ordered feature identity is enforced. I attacked all of these and landed nothing.

This verdict is conditional on one thing. F2 — audit independence — is unresolved, and the artifacts still describe themselves as independent verification while being self-attestation. As a *known and documented* limitation that is a warning. If those artifacts continue to claim independence they do not have, that alone is grounds to return to BLOCKED, because it misleads exactly the reader the seal exists to inform.

THE CHECK-BY-CHECK PROBE

Genuine acceptance, mutated one field at a time

```
$ # T1 - the accept path, never executed before R5-1 was fixed
T1 ACCEPT PATH: genuine acceptance -> PASSED all 10 gate checks, reached engine build. WORKS.

$ # then mutate exactly one field of that same genuine acceptance
check 6: validator SHA          -> BLOCKED (SMOKE_VALIDATOR_STALE)
check 7: violations > 0         -> BLOCKED (SMOKE_ACCEPTANCE_INVALID)
check 8: coverage < 100        -> BLOCKED (SMOKE_ACCEPTANCE_INVALID)
check 9: rows examined short    -> BLOCKED (SMOKE_ACCEPTANCE_INVALID)
check 9b: exact-equality False  -> reached engine build <== NOT ENFORCED
check 10: run_dir missing       -> BLOCKED (SMOKE_RUN_DIR_MISSING)

genuine acceptance restored
```

Checks 1–5 (status, study name, sealed composite, execution manifest composite, `deterministic_validation_verified`) were already proven in earlier rounds and use the same fail-closed `.get(...) != expected` form, which raises on a missing key rather than defaulting.

One new warning, three carried forward

WARNING R6-1

Producer and gate read the same config key with opposite defaults, leaving one check inert

backtests/nt_runtime/modes/collect.py:119-124 · scripts/validate_smoke.py:171 · studies/<s>/study.yaml

MECHANISM

Both sides key off `execution.observation_policy.required_source_relation`. The validator reads it as `obs_policy.get("required_source_relation", "equal")` — defaulting to strict. The gate reads it as `if obs_pol.get("required_source_relation") == "equal":` — requiring it to be set explicitly. The key is absent from `study.yaml` and resolves to `None` in the compiled contract, so `obs_pol` is `{}`, the condition is false, and the enclosed assertion never evaluates.

FAILURE

Demonstrated: an acceptance with `exact_timestamp_equality_verified: False` passes the gate. That combination is one the honest validator can never emit — it sets the flag as `bool(future_source_violations_count == 0)` — so the gate accepts a self-contradictory artifact.

SEVERITY

Warning, not critical. Check 7 enforces `future_source_violations_count != 0` unconditionally two lines above, and that is the substantive claim; the inert check is defence in depth on a derived flag. Recorded because the *pattern* — an optional config key silently deciding whether a safety check applies — is the same shape as R4-1's column-presence guard, and it defaults the wrong way.

FIX

Match the validator's default: `obs_pol.get("required_source_relation", "equal") == "equal"`. Better, add an internal-consistency check independent of config — `exact_timestamp_equality_verified` must equal `future_source_violations_count == 0` — so a contradictory acceptance is rejected on its own terms. Best, declare `observation_policy` explicitly in `study.yaml` so neither side has to default.

REGRESSION TEST

`test_gate_rejects_contradictory_acceptance` — `violations=0` with `exact_timestamp_equality_verified=False` must raise regardless of observation policy.

Preflight still carries a 17-minute market replay and still has no subprocess timeout

scripts/research_preflight.py:162-163 · pytest.ini · scripts/select_required_tests.py:58

STATUS

Unchanged. The pytest invocation is byte-identical — no `timeout=`, no `-m "not slow"`. `pytest.ini` declares no markers and no test carries `@pytest.mark.slow`. Measured last round: `test_nt_runner_collect.py` is 16 passed in 1008.03s (16m 48s); the full suite is 131 passed in ~17m 49s, and preflight runs all 14 files.

WHY IT STILL MATTERS

Nothing is failing — this is purely cost. The risk is second-order: a seventeen-minute "fast gate" invites `--skip-tests` as standing practice, which silently disables `CAUSAL_INVARIANTS` and reproduces the R4-3 condition by a different route. And with no timeout, a hung replay hangs preflight with no diagnostic.

FIX

Mark the end-to-end tests `slow`; run `-m "not slow"` in preflight and the full set at the pre-execution gate; pass `timeout=` to `subprocess.run` and treat expiry as `BLOCKED` with its own failure id.

Audit independence and agent permissions — open since rounds 2 and 1

scripts/preexec_audit_seal.py · .claude/agents/lookahead-auditor.md:3 ·
.claude/settings.local.json

F2 STATUS

`transcript_sha256` appears **zero times** in the seal — it is neither required nor checked. The parsing layer is genuinely strong now (structured summary, heading cross-check, v1 retired, seal re-derives), but the report is authored by whoever runs the flow, so the chain remains self-attestation with good formatting.

F13 STATUS

`lookahead-auditor` still declares `tools: [Read, Grep, Glob, Bash, Write]`. `settings.local.json` is unchanged since June and still has no `hooks` key, so `.claude/hooks/validate-results-triager-command.py` never executes. An auditor with shell access can issue its own status.

FIX

F2: either require a non-null `transcript_sha256` bound to an invocation record the orchestrator does not author, or rename the artifacts to `self_attested_*` and drop "independent" from `CLAUDE.md`, `AGENTS.md` and the seal's field names. F13: restrict the auditor to `[Read, Grep, Glob, Write]` with Write scoped to `audit/`, and register the existing hook under `hooks.PreToolUse`.

The gate is functional, and its checks are individually proven

backtests/nt_runtime/modes/collect.py

EVIDENCE

Symtable analysis confirms both names now resolve at module level. The accept path reaches engine construction on the genuine acceptance, and five of the six previously-unreachable checks each reject their own mutation with a distinct error. This is the first time any check beyond number five has executed.

NOTE ON METHOD

The round-5 recommendation was to pair every rejection test with an acceptance test. That is exactly what made this probe possible — and what would have caught the `NameError` before I did. It is worth adding to the suite rather than leaving as a manual procedure.

Three items, none of them mechanical defects

1. **R6-1** — one-line default alignment, plus the internal-consistency check that makes the gate independent of config.
2. **R5-2** — slow markers and a subprocess timeout. The longer this sits, the more likely `--skip-tests` becomes habit.
3. **F2** — the decision, not the code. Bind a transcript or rename the artifacts.

A closing note on the arc. Round 1 found five criticals and three landed bypasses; this round found none of either, and the difference is not that I looked less hard — it is that the controls now measure rather than assert. Each round's residue has been the same defect in a new place: a check that reports a favourable result it did not compute. R6-1 is that defect one more time, in its mildest form yet, caught only because a neighbouring check happens to cover it. **The convention that would retire the whole family is to make every safety check unconditional by default and require config to *relax* it, never to enable it.** Both R4-1 and R6-1 were checks that config had to switch on; the validator's own `get(..., "equal")` default is the correct pattern already present in the codebase.

Red team round 6, 2026-08-15 · study Gemini_clean_maturity_flip_rolling_5m_productivity · 53 sealed files · 131 tests passing · PREEXEC_AUDIT_SEAL_VALID · preflight CLEAR. All mutations executed against the live working tree; the genuine smoke_acceptance.json was backed up first and verified byte-identical after restore, with the 27-file modified set unchanged from session start.